

Likelihood and deviance

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Toy Data: Study hours and probability of passing

```
students <- tibble(
  hours = 1:6,
  pass = c(0, 0, 1, 0, 1, 1)
)
head(students)
```

STUDENT	HOURS	PASS
Adam	1	0
Bob	2	0
Carrie	3	1
David	4	0
Eve	5	1
Frank	6	1

```
mod <- glm(pass ~ hours, family = binomial, data = students)
summary(mod)
```

```
#> Coefficients:
#> Estimate Std. Error z value Pr(>|z|)
#> (Intercept) -4.2491    3.3878  -1.254    0.210
#> hours      1.2140    0.9126   1.330    0.183
#>
#> Null deviance: 8.3178  on 5  degrees of freedom
#> Residual deviance: 4.9560  on 4  degrees of freedom
#>
#> Number of Fisher Scoring iterations: 5
```

$$L_i = \begin{cases} p_i & y_i = 1 \\ 1 - p_i & y_i = 0 \end{cases}$$

- Each L_i is that student's **likelihood contribution**.

$$L = \prod_{i=1}^n L_i = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i}$$

PRACTICE Bob failed. The model's P(pass) = 0.139. Bob's likelihood contribution: (a) 0.861 (b) 0.139

Answer: 0.861

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Deviance

Deviance = $-2 \times \text{log-likelihood}$

Residual deviance = $-2 \times (-2.478) = 4.956$

- The main reason for the -2: **differences** in deviance between nested models follow a chi-squared distribution.

PRACTICE We care about deviance mainly because... (a) it lets us compare models (b) bigger = better fit

Answer: it lets us compare models

- Null deviance** comes from a model with no predictors.
- That model predicts 0.5 for everyone, because 3 of 6 students passed.

$$\ell_0 = 6 \times \ln(0.5) = 6 \times (-0.693) = -4.159$$

$$\text{Null deviance} = -2 \times (-4.159) = 8.318$$

- Null deviance** = deviance of the intercept-only model.
- Residual deviance** = deviance of the fitted model.
- Degrees of freedom = number of observations - number of parameters.

PRACTICE Null deviance is the deviance of... (a) the intercept-only model (b) the fitted model with predictors

Answer: the intercept-only model

PRACTICE Model A: deviance 50. Model B: deviance 40. Same data. Which model fits better? (a) Model B (b) Model A

Answer: Model B

$$L = .954 \times .861 \times .353 \times .353 \times .861 \times .954 = 0.084$$

- glm() chose the coefficients using **maximum likelihood estimation (MLE)**.
- R uses **Fisher scoring** to update the coefficients. "Number of Fisher Scoring iterations: 5" indicates five iterations.
- Multiplying many small numbers is difficult to work with. Logs turn that product into a sum.

$$\ell = \ln L = \ln(.954) + \ln(.861) + \ln(.353) + \ln(.353) + \ln(.861) + \ln(.954) = -2.478$$

$$\ell = -0.047 - 0.150 - 1.041 - 1.041 - 0.150 - 0.047 = -2.478$$

```
logLik(mod) #> 'log Lik.' -2.477987 (df=2)
```

PRACTICE Model X has likelihood 0.05. Model Y has 0.09. Which fits better? (a) Model Y (b) Model X

Answer: Model Y

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Learning objectives

YOU SHOULD NOW BE ABLE TO:

- ✓ Explain what a model's **likelihood** is.
- ✓ Explain why we use the **log** of the likelihood.
- ✓ Convert a log-likelihood into a **deviance** with $\times(-2)$. State which direction means a better fit.
- ✓ Identify **null deviance** and **residual deviance** in `summary()` output.
- ✓ State how degrees of freedom are calculated for deviance.

PRACTICE A model has log-likelihood -120. Its deviance is... (a) 240 (b) 120

Answer: 240

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FORMULA RECAP		
LIKELIHOOD	$L = \prod_i L_i$	Multiply each person's likelihood contribution.
LOG-LIKELIHOOD	$\ell = \ln L$	Add the logs of the likelihood contributions.
DEVIANCE	-2ℓ	-2 times the log-likelihood. Lower values indicate a better fit.
NULL DEVIANCE	deviance of $y \sim 1$	Deviance of the intercept-only model.
RESIDUAL DEVIANCE	deviance of your model	Deviance of the fitted model.

PRACTICE Which quantity is not reported by `summary()` for a glm? (a) log-likelihood (b) residual deviance

Answer: log-likelihood